Community-based Health Insurance and Health: Evidence from Senegal *

Ibou Dieye†

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Abstract

Community-based health insurance (CBHI) schemes have been promoted in low- and middle-income countries to extend financial protection to populations excluded from formal insurance, particularly in the large informal sector. Most evaluations find modest gains in utilization and out-of-pocket expenditures, but evidence on health outcomes is sparse and relies largely on observational comparisons. This paper provides the first nationwide, quasi-experimental evaluation of CBHI in Senegal, one of the few countries in sub-Saharan Africa to have pursued CBHI expansion as a central strategy for achieving universal health coverage. I exploit the staggered rollout of 676 mutual health organizations across departments between 1990 and 2019, linking administrative rollout data with eleven Demographic and Health Surveys and five Service Provision Assessment surveys, and apply difference-in-differences estimators. I trace effects along the causal chain from insurance coverage to financial protection, healthcare utilization, and health outcomes. CBHI increased maternal insurance coverage by 6.7 percentage points and reduced the likelihood of high out-of-pocket expenditures by 22.6 percentage points, though the coverage effect is sensitive to alternative specifications. Effects on healthcare utilization are limited, and I find no effects on pregnancy loss or neonatal mortality. Voluntary CBHI can reduce financial barriers, but low enrollment, weak provider contracting, and limited supply-side responsiveness constrain its effectiveness. Achieving universal health coverage will require more effective financing mechanisms and complementary investments in service quality and delivery systems.

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†Harvard University. ibrahimadieye@g.harvard.edu

1 Introduction

Universal health coverage (UHC) is a central goal of global health policy. It is enshrined in the 2030 Sustainable Development Goals, with Target 3.8 committing countries to ensure access to essential health services and financial protection (United Nations, 2015). Achieving UHC requires financing models that reach populations excluded from formal insurance, particularly in low- and middle-income countries (LMICs).

Progress has been uneven. In many LMICs, a high share of the population incurs impoverishing out-of-pocket (OOP) health spending (World Health Organization, 2023), and insurance coverage remains low. The informal sector, estimated to account for 80–90% of the workforce in many LMICs (International Labour Organization, 2018), is often excluded from contributory schemes. In response, governments and development partners have promoted community-based health insurance (CBHI), known in West Africa as mutual health organizations (MHOs). These voluntary arrangements pool community resources to provide financial protection and improve healthcare access (Chemouni, 2018; Alenda-Demoutiez and Boidin, 2019). CBHI has long been viewed as a promising approach for extending coverage where formal schemes are absent (Dror and Jacquier, 1999), but recent evidence has raised questions about its effectiveness and scalability.

A large body of research has examined the effects of CBHI on healthcare utilization and spending, finding modest improvements (Eze et al., 2023). Far fewer studies assess health impacts. Two studies in China and India report health improvements associated with CBHI (Wang et al., 2009; Aggarwal, 2010), but replication outside these settings has been limited. Two important gaps persist. First, most evaluations rely on observational comparisons between insured and uninsured individuals. Although many adjust for selection bias (Jütting, 2004; Smith and Sulzbach, 2008), concerns about endogeneity persist (Degroote et al., 2020). Second, few studies trace effects along the full causal chain from coverage to financial protection, utilization, and health outcomes.

The broader literature on public health insurance in LMICs offers a useful benchmark. Large, government-supported programs have sometimes achieved meaningful health gains: Conti and Ginja (2023) document a 10 percent reduction in infant mortality in poor municipalities following the rollout of Mexico's *Seguro Popular*, driven by increased hospital deliveries and treatment of perinatal conditions, and Gruber et al. (2023) find that China's New Cooperative Medical Scheme increased life expectancy by 2.7 percent. Yet even these programs showed mixed results across populations and settings (Dupas and Jain, 2026). Both combined insurance expansion with direct supply-side investments and achieved substantially higher enrollment than voluntary CBHI schemes. Whether smaller, community-managed schemes can replicate such effects remains an open question.

Senegal provides a useful setting to address these gaps. Unlike many countries where CBHI remains small-scale or fragmented, Senegal launched a nationwide expansion of MHOs as part of its UHC strategy in 2013 (Daff, 2021). Between 2013 and 2018, the number of CBHI beneficiaries grew from 421,670 to over 3 million (ANACMU, 2019a). During 2015–2018, the government allocated 7.4 billion CFA francs (about USD 12.3 million) in premium subsidies to MHOs, nearly 10% of all subsidies for health utilization in that period (Cour des Comptes, 2019). Even as evidence on CBHI effectiveness has grown more skeptical, Senegal has continued to invest in and expand its CBHI model, making it one of the few countries in sub-Saharan Africa to have scaled community-based insurance to national coverage. This persistence makes rigorous evaluation not just academically useful but directly policy-relevant. Despite this investment, no prior study has established a causal link between CBHI coverage and health outcomes in Senegal. Existing evaluations have been cross-sectional and geographically limited. Jütting (2004) found reduced hospital expenditures among insured households in Thiès but noted lower enrollment among the poor. Chankova et al. (2008) documented modest protection against catastrophic hospitalization costs in Ghana, Mali, and Senegal. Smith and Sulzbach (2008) reported higher maternal healthcare utilization conditional on coverage in underserved areas. More recent evaluations found benefits concentrated among non-poor enrollees (Ly et al., 2022; Bousmah et al., 2022).

This study provides the first nationwide, quasi-experimental evaluation of Senegal's CBHI program. I assemble and link data from eleven rounds of the Demographic and Health Surveys (DHS), five Service Provision Assessment (SPA) surveys, and newly compiled administrative records documenting the rollout of 676 MHOs across all Senegalese departments.[1] I exploit staggered CBHI implementation from 1990 to 2019 using difference-in-differences methods designed for staggered adoption settings. I focus on maternal, reproductive, and neonatal health, domains central to CBHI benefit packages (Rouyard et al., 2022) and well captured in survey data (Croke et al., 2020). My analysis traces effects along the causal chain from insurance coverage to financial protection, healthcare utilization, and health outcomes. I examine heterogeneity by urban-rural setting and conduct a range of robustness checks, including honest difference-in-differences bounds, placebo tests, and checks addressing potential exposure misclassification.

CBHI increased insurance coverage for maternal care by 6.7 percentage points (95% CI: 2.4, 11.1) and reduced the likelihood of top-quintile OOP expenditures by 22.6 percentage points (95% CI: –36.0, –9.1), though the evidence on coverage is sensitive to alternative

[1] Departments are Senegal's second-level administrative subdivisions, comparable to districts in other countries.

specifications. Effects on healthcare utilization are limited, with small and generally not statistically significant estimates across most measures. I find no significant effects on pregnancy loss or neonatal mortality. Results are broadly robust to a range of checks addressing parallel trends violations, exposure misclassification, and alternative treatment definitions, though precision declines in specifications that require aggregation or impose additional restrictions. Heterogeneity analysis does not reveal a consistent pattern across urban and rural areas.

Unlike the health gains documented for larger, government-backed programs in Mexico and China, voluntary CBHI in Senegal did not translate coverage gains into measurable health improvements. This is consistent with evidence that voluntary schemes face structural barriers, including low enrollment and weak provider contracting (Rouyard et al., 2022), and that even heavily subsidized schemes often fail to achieve mass coverage (Dupas and Jain, 2026). National audits similarly document billing irregularities among contracted providers (Cour des Comptes, 2019).

My findings illustrate both the promise and the limits of CBHI as a pathway to UHC. Gains in coverage and financial protection suggest that community-based schemes can reduce cost barriers. Yet low enrollment, modest utilization effects, and the absence of health outcome improvements highlight the difficulty of translating insurance coverage into better population health without concurrent investments in service quality and stronger delivery systems (Dupas and Jain, 2026).

This study contributes to two bodies of literature. The broader literature on public health insurance in LMICs finds that insurance expansions generally increase financial protection and utilization, but evidence on health impacts remains limited and context-dependent (Dupas and Jain, 2026). The narrower CBHI literature largely echoes this pattern, with modest gains in utilization and financial protection but little evidence of health improvements (Eze et al., 2023). A recurring challenge is the reliance on observational designs that cannot credibly isolate causal effects. I add to both literatures with the first nationwide, quasi-experimental evaluation of CBHI in a sub-Saharan African country, combining multiple nationally representative data sources with administrative rollout records and modern difference-in-differences methods. The evidence spans the full causal chain from coverage to health outcomes, providing a more complete picture of what voluntary CBHI can and cannot achieve at scale.

The remainder of the paper is organized as follows. Section 2 provides institutional background on CBHI in Senegal. Section 3 describes the data. Section 4 outlines the empirical strategy. Section 5 presents results. Section 6 concludes.

2 Institutional setting

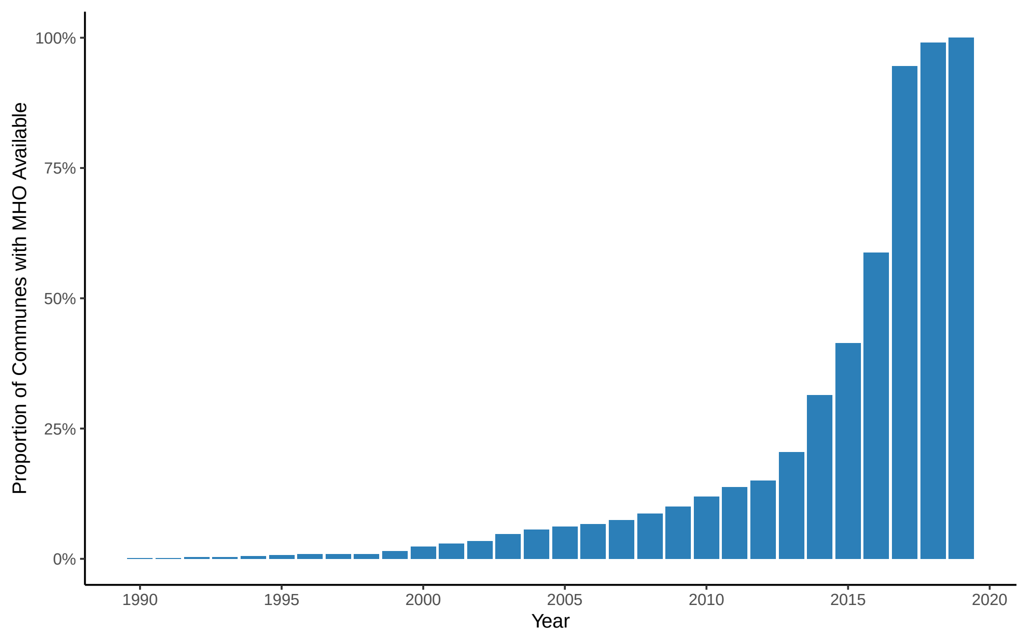
2.1 CBHI rollout in Senegal

Senegal's CBHI model began in 1989 with the establishment of the first mutual health organization (MHO) in Fandène (Thiès region), inspired by mutualist practices in France (Jütting, 2004; Alenda-Demoutiez and Boidin, 2019). By 2000, sixteen MHOs were active in the region, covering roughly 30,000 individuals (Jütting, 2004). These early schemes were organized at the community level, with volunteer boards overseeing enrollment, premium collection, and negotiations with local providers.

The Ministry of Health created a dedicated office to promote MHOs in 1998, and by 2008 CBHI had been incorporated into the national strategy for financial risk protection (Rouyard et al., 2022). A major policy shift came in 2013, when CBHI was adopted as a central component of Senegal's UHC strategy. In 2015, the government established ANACMU to standardize and scale implementation nationwide (Daff et al., 2020).

Figure 1 shows the pace of rollout. In 2012, 15% of communes had access to CBHI, increasing to 60% by 2016. By 2019, coverage extended to all communes.

Figure 1. Cumulative rollout of MHOs across communes



Note: This figure shows the cumulative proportion of communes with at least one MHO by year. The data are collapsed at the commune level using the earliest year in which an MHO was active. *Abbreviations*: MHO = mutual health organization.

Under ANACMU oversight, MHOs adopted a uniform benefit package, eligibility criteria, and premium structure. Households paid annual premiums of roughly 7,000 CFA francs (about USD 12) per person, with the government subsidizing 50% for the general population and fully covering premiums for vulnerable groups such as low-income households and persons with disabilities (Daff et al., 2020; Rouyard et al., 2022). The package covered outpatient care, maternal and newborn services, and hospitalization, subsidizing 80% of user fees and generics and 50% of branded pharmaceuticals. Benefits were not portable, and enrollees were covered only for services at facilities contracted by their local MHO. Contracted networks generally included primary, secondary, and tertiary facilities within the region, with referrals required for hospital coverage. Facilities were reimbursed retrospectively on a fee-for-service basis at rates negotiated with the MHOs. To expand risk pooling for high-cost services, MHOs formed unions at the departmental level (Deville et al., 2018; Rouyard et al., 2022).

To address the limitations of small-scale risk pools and volunteer management, the government piloted Departmental Health Insurance Units (UDAMs) in Koungheul and Foundiougne between 2014 and 2017 (Daff et al., 2020). These professionally managed schemes operated at the department level. Senegal began scaling UDAMs in 2020 (Ridde et al., 2022). Because data on UDAMs remain limited and my analysis covers the pre-2020 period, I exclude the two pilot departments and focus on the commune-level CBHI model implemented across the remaining 43 departments.

2.2 CBHI access and maternal health

CBHI may influence health outcomes through several channels. By reducing out-of-pocket costs, it lowers financial barriers to care and can encourage timely use of antenatal, delivery, and postnatal services (Smith and Sulzbach, 2008). Enrollment may also foster engagement with the health system, supporting preventive care and continuity of treatment. On the supply side, contractual arrangements between MHOs and providers may affect provider behavior and care quality.

In practice, however, these channels do not operate cleanly. Providers may respond to insurance by raising prices, delivering unnecessary care, or creating uncertainty about benefit delivery, thereby undermining coverage gains (Das and Do, 2023). Take-up may also be limited: even heavily subsidized schemes often fail to achieve mass enrollment, constraining the scope for population-level health effects (Dupas and Jain, 2026). In Senegal specifically, operational weaknesses including weak provider contracting and low member retention further limit program reach (Rouyard et al., 2022).

CBHI schemes may also generate spillovers. Outreach and sensitization efforts can raise community awareness of health risks and the value of timely care-seeking, even among non-enrolled individuals.

I exploit staggered MHO rollout across departments to estimate causal effects on insurance coverage, financial protection, healthcare utilization, and maternal and neonatal health outcomes. By focusing on community-managed MHOs and excluding departmental UDAMs, my analysis isolates the impacts of Senegal's flagship CBHI model during its nationwide expansion.

3 Data

3.1 Data sources

This study combines four primary data sources: administrative records from Senegal's National Agency for Universal Health Coverage (ANACMU), nationally representative household data from the Demographic and Health Surveys (DHS), healthcare facility data from the Service Provision Assessment (SPA) surveys, and population estimates from the National Agency of Statistics and Demography (ANSD). Table 1 summarizes the years covered and the key measures drawn from each source. DHS surveys, which span the full study period (1992–2019), provide information on insurance coverage, healthcare utilization, and health outcomes. SPA surveys, available for 2012–2017, complement these with facility-level data on insurance and out-of-pocket costs.

3.1.1 ANACMU

I use administrative records from ANACMU to determine the timing of CBHI rollout. The dataset documents 676 MHOs across 552 communes as of 2019, of which 666 are community-based and 10 are professional MHOs targeting occupational groups. I exclude the latter.

For most MHOs (601 of 666, or 91%), the dataset records the year of operational launch, which I use as the exposure date. For 49 MHOs lacking launch years, I substitute the creation year; the average lag between creation and launch is 0.8 years. For 16 MHOs with no date information (8 in Tivaouane, 2 in Thiès, 3 in Mbour, and 3 in Fatick), I impute rollout dates using the commune average when available, or the department-level average otherwise. I conduct robustness checks excluding departments with any imputed launch years. I also drop 26 MHOs in Foundiougne and Koungheul, which piloted departmental CBHI units (UDAMs) from 2014.

3.1.2 DHS

I use eleven DHS rounds conducted in Senegal between 1992 and 2019. These nationally representative surveys provide data on health, healthcare utilization, and demographic characteristics (Corsi et al., 2012; Demographic and Health Surveys Program, 2022). I draw on three modules.

Birth history module. Retrospective data on all children ever born to women aged 15–49, including survival status and maternal care. The unit of observation is the birth.

Women's interview module. Individual-level data on family planning, pregnancy, insurance, healthcare utilization, and socioeconomic variables. The unit of observation is the woman.

Contraceptive calendar. Monthly histories of reproductive events and contraceptive use over the 5–7 years preceding the interview, allowing reconstruction of pregnancy episodes and pregnancy loss.

All DHS rounds include georeferenced cluster coordinates, which I use to assign CBHI exposure based on departmental rollout (see Section 3.2).

3.1.3 SPA

Five SPA surveys conducted between 2012 and 2017 offer nationally representative or near-census assessments of healthcare facility capacity (Kruk et al., 2017; Demographic and Health Surveys Program, 2022). I rely on exit interviews with women seeking antenatal care (ANC) and family planning (FP) services, which provide individual-level data on insurance coverage and out-of-pocket costs. These complement the DHS by offering direct measures of financial protection during the 2012–2017 period.

3.1.4 ANSD population estimates

Commune-level population estimates are drawn from the 2002 and 2013 censuses and official projections for 2014–2019 (ANSD, 2019). Because administrative redistricting increased the number of communes over time, some communes listed after 2013 did not exist in 2002.

To generate consistent annual commune-level estimates from 1990 to 2019, I align commune identifiers across years and interpolate gaps using natural splines (Siegel and Swanson, 2004). Validation exercises show minimal prediction error relative to observed census counts (Appendix Figure A1). These estimates are used to construct CBHI exposure variables.

Table 1. Summary of data sources

Data source	Years	Survey modules	Measures extracted
ANACMU	1990–2019	–	MHO initiation and launch year
DHS	1992–2019	• Women's interview • Birth history • Contraceptive calendar • Geocoordinates	• Insurance coverage • FP uptake • Antenatal care • Delivery care • Pregnancy outcomes • Neonatal health
SPA	2012–2017	• Exit interviews (ANC and FP) • Facility audits and interviews • Geocoordinates	• Insurance coverage • OOP costs
ANSD	2002, 2013–2019	–	Commune-level population data

Note: This table summarizes the data sources used in the analysis, including survey years, modules, and key measures extracted. *Abbreviations*: ANC = antenatal care; ANACMU = National Agency for Universal Health Coverage; ANSD = National Agency of Statistics and Demography of Senegal; DHS = Demographic and Health Surveys; FP = family planning; MHO = mutual health organization; OOP = out-of-pocket; SPA = Service Provision Assessment.

3.2 Exposure measures

I construct CBHI exposure measures by combining administrative rollout data with GPS-located DHS and SPA surveys. DHS cluster coordinates are randomly displaced (up to 2 km in urban areas, 5 km in rural areas, and up to 10 km for 1% of rural clusters) (Demographic and Health Surveys Program, 2024), making commune-level assignment unreliable. I therefore assign exposure at the department level, where displacement is constrained for most survey rounds. SPA coordinates are not displaced, so department-level assignment is exact for facility analyses.

DHS clusters are mapped to departments using contemporaneous shapefiles from Senegal's National Agency for Land Use Planning (ANAT) (United Nations, 2024) and the `sf` package in R (Pebesma, 2018). Since 2009, DHS displacement has been restricted within departmental boundaries. I re-estimate results restricting to post-2009 rounds as a robustness check.

Within each department, MHOs launched in staggered years. To approximate when the typical resident first gained access, I compute a population-weighted average launch year, using commune population in the year prior to each launch. This ensures weights are not influenced by rollout itself. Table 2 illustrates the calculation for Gossas, where six MHOs launched between 2014 and 2017, yielding a department exposure year of 2016.

Table 2. MHO launch years and pre-launch population in Gossas

Department	**Commune**	**MHO launch year**	**Population (year−1)**
Gossas	Gossas	2014	12,698
Gossas	Ndiene Lagane	2016	12,652
Gossas	Colobane	2017	20,172
Gossas	Mbar	2017	30,077
Gossas	Ouadior	2017	11,916
Gossas	Patar Lia	2017	16,221

Formally, letting y_i denote the MHO launch year in commune i and w_i its pre-launch population, the weighted average launch year for department d is:

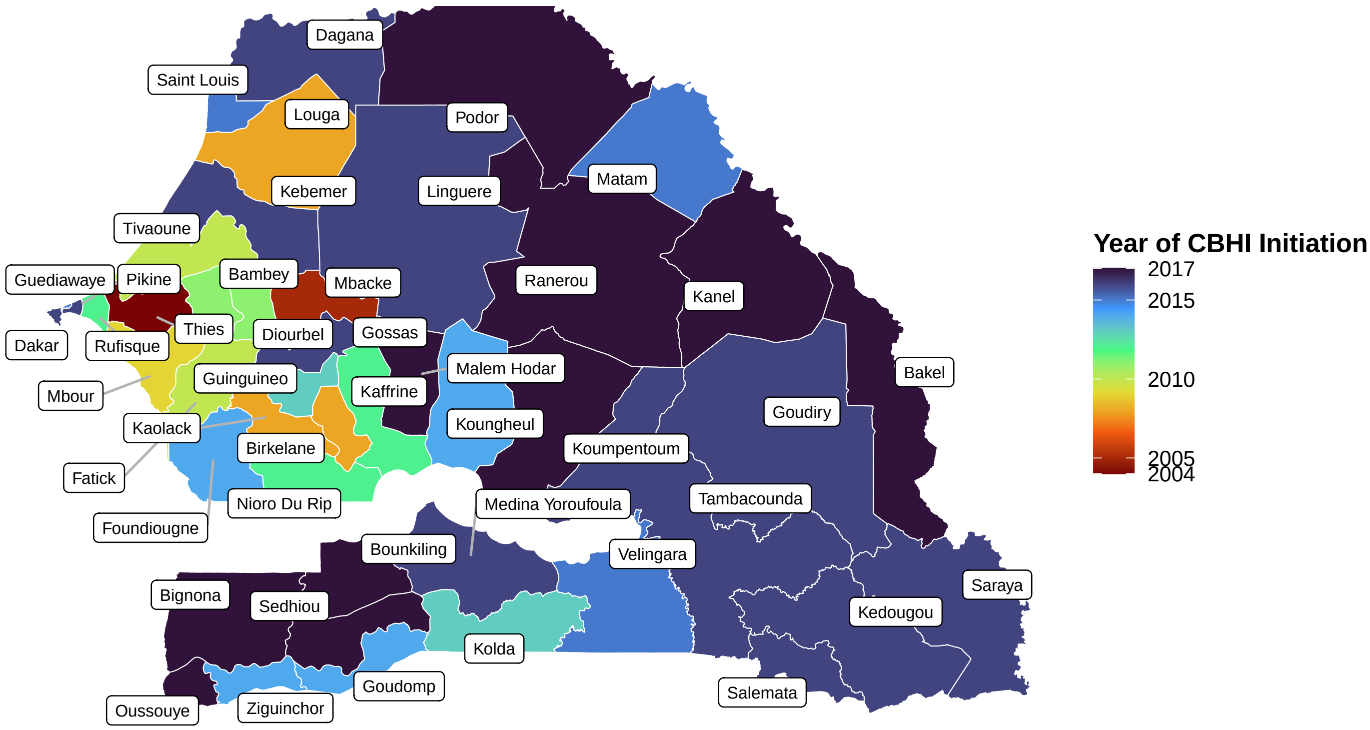
$$\text{Exposure}_d = \left\lfloor \frac{\sum_{i \in d} y_i \cdot w_i}{\sum_{i \in d} w_i} \right\rfloor,$$

where the floor operator assigns the latest full year prior to when the typical resident gained access to CBHI.

This measure improves on a binary indicator based on the first MHO launch but has limitations. It does not capture within-department intensity or geographic spread, may introduce measurement error that attenuates estimates, and could contaminate pre-treatment periods due to gradual rollout. I address these concerns through robustness checks in Section 5.3.

Figure 2 shows variation in rollout timing across Senegal's 45 departments. Thiès was the earliest adopter, with exposure beginning in 2004, while ten departments, including Bakel, Podor, and Kanel, did not gain access until 2017.

Figure 2. Geographic variation in CBHI rollout timing



Note: This figure displays the timing of CBHI exposure across Senegal's 45 departments as of 2019. Exposure timing is defined as the population-weighted average MHO launch year across communes within each department.

3.3 Outcome measures

I group outcomes into four categories reflecting the causal chain from insurance coverage to health: first-stage (insurance uptake), healthcare costs, healthcare utilization, and health outcomes.

3.3.1 First-stage and healthcare cost outcomes

The first-stage outcome is a binary indicator for whether a woman reported having insurance that covered the cost of ANC or FP services at the time of her facility visit. Healthcare costs are measured by a binary indicator equal to one if a woman's OOP expenditure falls within the top quintile of the survey-year cost distribution. This relative measure accounts for inflation and secular changes in user fees. Both outcomes are derived from SPA exit interviews.

Because these outcomes are only observed among women who sought care, estimates capture effects among care users rather than the general population and may be upwardly biased if insurance also increases care-seeking.

3.3.2 Healthcare utilization outcomes

I construct three indices of care quality and intensity following Anderson (2008), to improve statistical power and reduce multiple testing.

ANC visit index. Share of three recommended components received: (i) at least four ANC visits, (ii) initiation of ANC by the end of the first trimester, and (iii) care from a skilled provider (doctor, nurse, midwife, or auxiliary midwife).

ANC process index. Share of six recommended ANC services received: blood pressure measurement, blood sample collection, urine sample collection, receipt of iron supplements, at least two doses of tetanus toxoid vaccine, and at least three doses of intermittent preventive treatment for malaria during pregnancy.

Delivery care index. Share of two recommended delivery components: facility delivery and skilled provider attendance.

Each index ranges from 0 to 1. The components align with WHO guidelines for a positive pregnancy experience (World Health Organization, 2016).

3.3.3 Health outcomes

I focus on two health outcomes measured in the DHS.

Pregnancy loss. Any pregnancy ending in a non-live birth, including miscarriage, stillbirth, or abortion, as recorded in the contraceptive calendar.

Neonatal death. Death of a live-born infant within the first 28 days, using the DHS birth history module.

I restrict attention to neonatal outcomes because children under five were eligible for overlapping free care programs that could confound CBHI effects.

3.4 Additional measures

I draw on demographic and socioeconomic characteristics from the DHS and SPA surveys, including age, education, household wealth quintiles, and marital status. These describe the study population, serve as covariates in robustness checks, and provide placebo outcomes. All variables are harmonized across survey rounds to ensure comparability over time.

3.5 Summary statistics

Tables 3, 4, and 5 present summary statistics for the DHS women's interview, DHS birth history, and SPA exit interview samples, respectively. Statistics are shown overall and stratified by pre- and post-CBHI exposure periods.

In both the DHS and SPA samples, the average woman is about 28 years old, married, has limited formal education, is not formally employed, and belongs to the bottom two household wealth quintiles (Tables 3 and 5). In the birth history sample, the average age at first birth is around 19 years (Table 4).

Women surveyed before and after CBHI rollout are broadly similar across all samples. Education levels and wealth quintiles are somewhat higher in the post-rollout period, likely reflecting broader socioeconomic changes over time. My difference-in-differences framework accounts for these differences.

Table 3. Summary statistics of the DHS women's sample, by survey timing

Characteristic	**Overall** N = 113,542	**Survey before CBHI** N = 39,891	**Survey after CBHI** N = 73,651
Age	27.90 (9.34)	27.88 (9.35)	27.91 (9.33)
Education level: no education	58%	63%	55%
Education level: primary	21%	21%	21%
Education level: secondary	20%	15%	22%
Education level: higher than secondary	1.5%	1.0%	1.8%
Married	68%	69%	68%
Employed	44%	43%	44%
Mother age at 1st birth	19.57 (4.03)	19.03 (3.81)	19.88 (4.11)
Education level, husband: no education	76%	78%	75%
Education level, husband: primary	11%	10%	11%
Education level, husband: secondary	9.3%	8.8%	9.7%
Education level, husband: higher than secondary	3.4%	2.9%	3.7%
Age, husband	43.49 (13.92)	43.13 (13.48)	43.62 (14.07)
Residence type: urban	40%	39%	40%
Number of household members	13.67 (8.39)	13.40 (8.41)	13.81 (8.38)
Household wealth quintile: poorest	23%	25%	22%
Household wealth quintile: poorer	23%	25%	22%
Household wealth quintile: middle	23%	24%	23%
Household wealth quintile: richer	18%	15%	19%
Household wealth quintile: richest	14%	11%	15%

Note: Characteristics of women interviewed in the DHS, overall and by timing of interview relative to CBHI exposure. Means and standard deviations (SD) are reported for continuous variables; percentages for categorical variables. *Abbreviations*: CBHI = community-based health insurance.

Table 4. Summary statistics of the DHS birth sample, by pregnancy timing

Characteristic	**Overall** N = 306,039	**Pregnancy before CBHI** N = 228,882	**Pregnancy after CBHI** N = 77,157
Sex of baby: female	49%	49%	49%
Sex of baby: male	51%	51%	51%
Birth order	3.35 (2.24)	3.28 (2.22)	3.53 (2.29)
Twin birth	3.1%	2.8%	3.9%
Age, mother	35.30 (7.84)	36.23 (7.73)	32.56 (7.50)
Education level, mother: no education	78%	79%	73%
Education level, mother: primary	16%	15%	19%
Education level, mother: secondary	5.8%	5.0%	7.9%
Education level, mother: higher than secondary	0.5%	0.3%	0.8%
Education level, father: no education	81%	82%	79%
Education level, father: primary	9.7%	9.4%	11%
Education level, father: secondary	7.0%	6.7%	7.8%
Education level, father: higher than secondary	2.2%	2.0%	2.8%
Age, father	48.30 (13.00)	49.66 (12.82)	45.15 (12.86)
Residence type: urban	31%	32%	30%
Number of household members	13.90 (8.40)	13.75 (8.51)	14.34 (8.03)
Household wealth quintile: poorest	30%	33%	23%
Household wealth quintile: poorer	26%	27%	25%
Household wealth quintile: middle	21%	21%	23%
Household wealth quintile: richer	14%	13%	17%
Household wealth quintile: richest	8.6%	7.7%	11%

Note: Characteristics of women in the DHS birth history sample, pooling across all births, overall and by timing of pregnancy relative to CBHI exposure. Means and standard deviations (SD) are reported for continuous variables; percentages for categorical variables. *Abbreviations*: CBHI = community-based health insurance.

Table 5. Summary statistics of the SPA sample, by survey timing

Characteristic	**Overall** N = 4,462	**Survey before CBHI** N = 1,356	**Survey after CBHI** N = 3,106
Age	27.83 (7.16)	28.08 (7.28)	27.72 (7.11)
Education level: no education	55%	55%	54%
Education level: primary	26%	27%	26%
Education level: secondary	15%	14%	16%
Education level: higher than secondary	4.0%	3.9%	4.1%
Location type: urban	47%	43%	48%
Facility type: health post	66%	66%	66%
Facility type: health center	25%	27%	24%
Facility type: hospital	8.4%	6.9%	9.0%
Facility ownership: public	94%	95%	93%
Facility ownership: non-profit	3.2%	1.8%	3.8%
Facility ownership: private	2.3%	2.1%	2.3%
Facility ownership: faith-based	1.0%	0.7%	1.1%
Service type: antenatal care	45%	44%	46%
Service type: family planning	55%	56%	54%

Note: Characteristics of women interviewed in the SPA exit survey, overall and by timing of survey relative to CBHI exposure. Age is reported as mean and standard deviation. All other variables are reported as percentages. *Abbreviations*: CBHI = community-based health insurance.

Appendix Table B1 presents SPA statistics stratified by insurance status. Insured women tend to be older, more educated, and more likely to live in urban areas, consistent with prior evidence that CBHI enrollees are typically of higher socioeconomic status than non-enrollees (Jütting, 2004; Conde et al., 2022). Such women may also exhibit greater health-seeking behavior, suggesting adverse selection into insurance. As a result, simple comparisons between insured and uninsured groups may overstate insurance benefits, even after adjusting for observables. I use a difference-in-differences design to address this concern.

4 Empirical strategy

4.1 Baseline specification

I estimate the impact of CBHI using a difference-in-differences (DiD) framework that exploits variation in the timing of MHO rollout across departments. My baseline two-way fixed effects (TWFE) specification is:

$$Y_{idts} = \beta_0 + \beta_1 \cdot MHO_d + \beta_2 \cdot (\text{post}_{dt} \times MHO_d) + X_i + \delta_d + \phi_t + \gamma_s + \epsilon_{idts}, \tag{1}$$

where Y_{idts} is the outcome for individual i in department d, year t, and survey round s. MHO_d equals one if department d ever receives treatment, and post_{dt} equals one if the observation occurs in or after the first year of CBHI exposure. The vector X_i includes individual-level covariates. Department fixed effects (δ_d) absorb time-invariant differences across departments, year fixed effects (ϕ_t) capture national trends, and survey-round fixed effects (γ_s) account for survey-specific shocks.[2] Standard errors are clustered at the department level.

This specification compares outcomes within departments before and after CBHI rollout, using not-yet-treated departments as controls. The identifying assumption is that, absent treatment, treated and not-yet-treated departments would have followed parallel trends. Fixed effects address common national shocks and persistent cross-department differences, but do not rule out unobserved shocks that vary over time and across space. A further concern is reverse causality if rollout decisions responded to recent or anticipated outcome changes. Historical accounts suggest that rollout timing reflected social capital, logistical capacity, donor support, and political factors rather than local health trends (Alenda-Demoutiez and Boidin, 2019; Wood, 2023), supporting the plausibility of the identifying assumption. I assess it formally using event-study tests in Section 5.

[2]In practice, survey year fixed effects are dropped in estimation due to collinearity with year fixed effects.

For individual-level DHS analyses and SPA first-stage and cost outcomes, the time variable corresponds to the survey year. For DHS birth history and contraceptive calendar modules, it refers to the year of pregnancy onset. In the contraceptive calendar, pregnancy start month and year are observed directly. In the birth history module, pregnancy onset is approximated as nine months before the reported birth year. In birth-level analyses, I assume women resided in the same department at childbirth as at survey, consistent with prior work (Croke et al., 2020).

4.2 Estimation strategy

TWFE models may yield biased estimates in settings with staggered adoption and heterogeneous treatment effects (Goodman-Bacon, 2021). My main analysis therefore uses the doubly robust DiD estimator proposed by Callaway and Sant'Anna (2021), which avoids negative weighting and allows for treatment effect heterogeneity across cohorts and over time.

I estimate average treatment effects (ATEs) for the overall sample and separately for urban and rural strata, using not-yet-treated comparison groups within each stratum. Dynamic effects are examined with an event-study specification:

$$Y_{idts} = \alpha + \sum_{k=-5}^{5} \beta_k \cdot \mathbf{1}\{t - T_d = k\} + X_i + \delta_d + \phi_t + \gamma_s + \epsilon_{idts}, \tag{2}$$

where T_d is the year of CBHI rollout in department d and $\mathbf{1}\{t - T_d = k\}$ indicates k years relative to rollout, with $k = -1$ omitted. Department, year, and survey-round fixed effects are included, and standard errors are clustered at the department level. Leads and lags beyond five years are pooled into $k = -5$ and $k = 5$ bins to improve stability.

Following standard best practices (Baker et al., 2025), my main specification assumes unconditional parallel trends, which is supported by the event studies. I examine robustness to alternative specifications in Section 5.3.

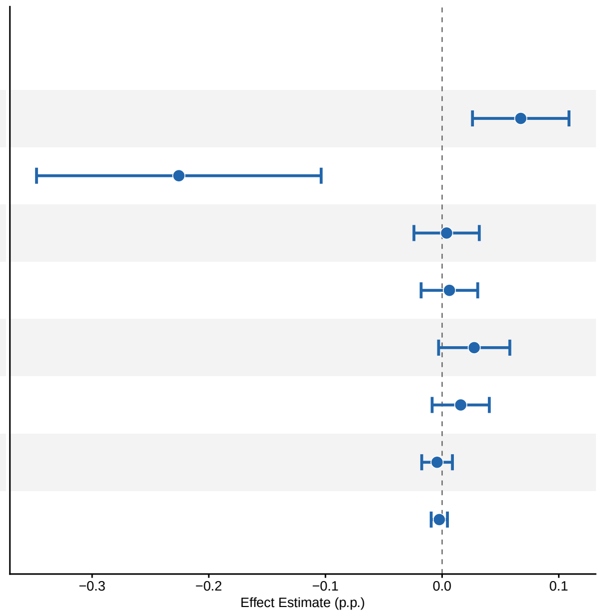
5 Results

5.1 Main results

Figure 3 presents ATE estimates from the baseline specification for the overall sample. Event study plots are shown in Figures 4–6.

Figure 3. ATE estimates

Outcome	Effect (95% CI)	Mean	N
Any Insurance Coverage	0.067 (0.026, 0.109)*	0.076	1,614
OOP Cost in Q5	−0.226 (−0.348, −0.104)*	0.187	1,612
ANC Visits Index	0.004 (−0.024, 0.032)	0.707	56,840
ANC Process Index	0.006 (−0.018, 0.031)	0.737	56,771
Delivery Care Index	0.028 (−0.003, 0.058)	0.820	79,681
Secondary/Tertiary Facility Delivery	0.016 (−0.009, 0.040)	0.284	79,660
Pregnancy Loss	−0.004 (−0.017, 0.009)	0.100	48,559
Neonatal Death	−0.002 (−0.009, 0.005)	0.028	288,420



Note: This figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; DHS = Demographic and Health Surveys; FP = family planning; OOP = out-of-pocket; SPA = Service Provision Assessment.

5.1.1 First stage: insurance coverage

Figure 4 shows broadly flat pre-trends for insurance coverage, though one pre-treatment period ($t = -3$) shows a significant negative deviation. I assess robustness to potential violations of the parallel trends assumption in Section 5.3.

CBHI increases insurance coverage for ANC or family planning services by 6.7 percentage points (95% CI: 2.6, 10.9). This represents an 88% increase relative to the baseline coverage rate of 7.6%, which aligns with prior estimates for informal sector populations in Senegal (Schwettmann, 2022). The effect is consistent with administrative data showing national coverage roughly doubled from about 20% to 50% between 2013 and 2019 (ANACMU, 2019b).

5.1.2 Healthcare costs

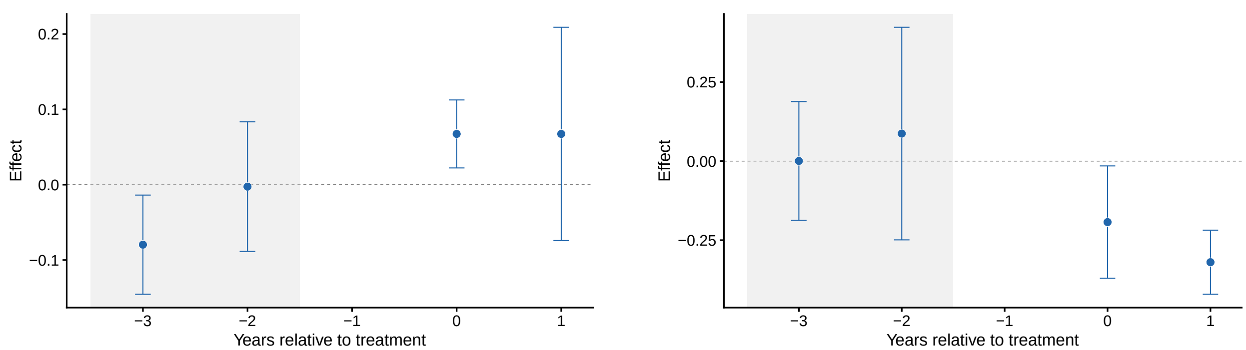
Pre-treatment trends for the cost outcome are flat (Figure 4).

CBHI reduces the likelihood of incurring high OOP costs by 22.6 percentage points (95% CI: –34.8, –10.4). These findings are consistent with earlier evidence from Senegal. Jütting (2004) found that MHO members in Thiès paid less than half the hospital costs of non-members. Chankova et al. (2008) documented that MHO membership reduced catastrophic hospitalization costs in Ghana, Mali, and Senegal, but not for outpatient care. My SPA sample does not distinguish inpatient from outpatient visits, though 8.4% of respondents reported visiting a hospital and 25% a health center.

Equity concerns temper this finding. Ly et al. (2022) found that CBHI reduced catastrophic expenditures among non-poor members but not among the poorest. Bousmah et al. (2022) reported similar patterns for fully subsidized members. Insured women in my SPA sample are older, wealthier, and more urban (Table B1), consistent with adverse selection into CBHI among the non-poor. Because the SPA data capture only women who sought care, and insured women may be more likely to seek care, the estimated reduction in OOP costs may overstate population-wide effects.

Figure 4. Event study estimates: first stage and healthcare costs

(a) First stage (b) High OOP costs



Note: Event-study estimates using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. Both outcomes are estimated at the woman level using SPA exit interview data. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. *Abbreviations*: ANC = antenatal care; FP = family planning; OOP = out-of-pocket; SPA = Service Provision Assessment.

5.1.3 Healthcare utilization

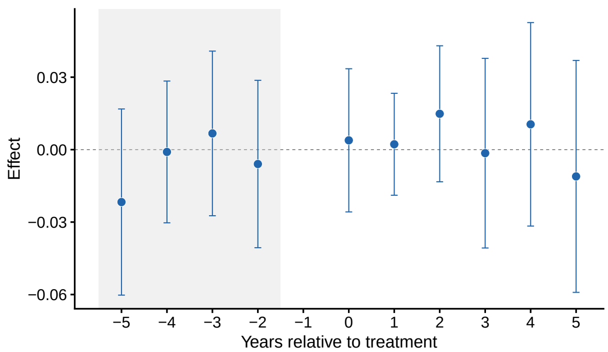
Pre-treatment trends are flat for all utilization outcomes (Figure 5).

CBHI increases the delivery care index by 2.8 percentage points (95% CI: –0.3, 5.8), though this estimate is not statistically significant at conventional levels. Effects on ANC visits and ANC process quality are small and also not statistically significant. These patterns echo prior studies in Senegal: Jütting (2004) and Smith and Sulzbach (2008) found that CBHI increased institutional delivery but had weaker effects on prenatal care. Chankova et al. (2008) similarly found limited effects on ANC visits across Ghana, Mali, and Senegal. More recent evaluations of subsidized CBHI in Senegal also found modest impacts beyond delivery care (Bousmah et al., 2022; Ly et al., 2022).

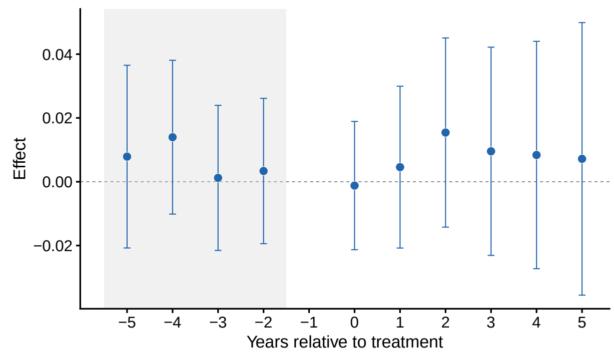
CBHI thus has limited effects across all dimensions of maternal care utilization.

Figure 5. Event study estimates: healthcare utilization

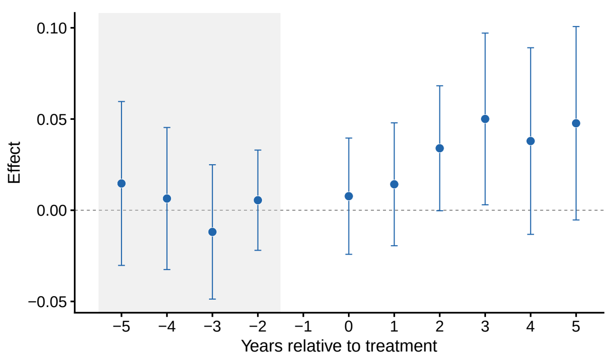
(a) ANC visits index



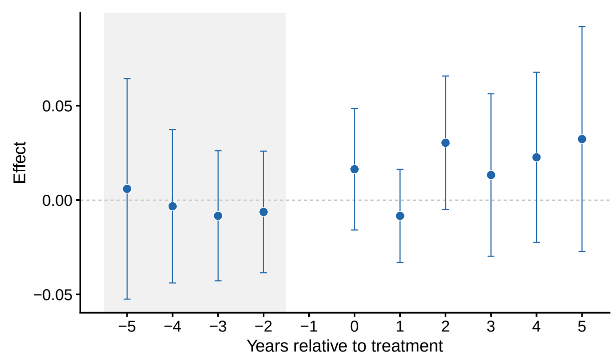
(b) ANC process index



(c) Delivery care index



(d) Delivery in a secondary or tertiary facility



Note: Event-study estimates using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. All outcomes are derived from DHS birth histories and estimated at the birth level. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. Other outcomes are binary. *Abbreviations*: ANC = antenatal care; DHS = Demographic and Health Surveys.

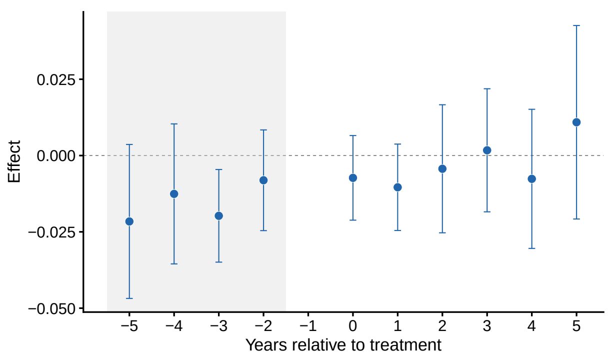
5.1.4 Health outcomes

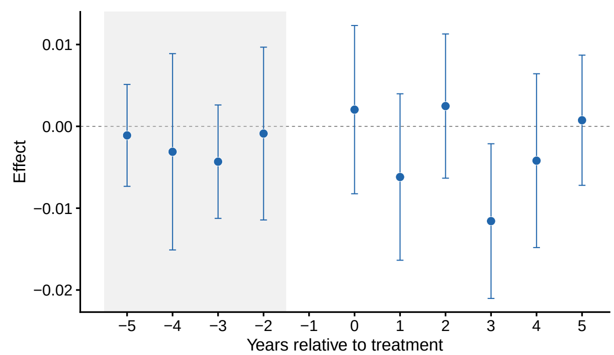
Figure 6 presents event-study estimates for health outcomes. For pregnancy loss, pre-trends are broadly flat but show one significant deviation at $t = -3$; I assess robustness to this in

Section 5.3. Pre-treatment trends are flat for neonatal mortality.

CBHI has no significant effect on pregnancy loss or neonatal mortality. These null results are consistent with the broader causal chain. CBHI increased coverage and reduced financial barriers, with modest gains in delivery care, but these did not translate into improvements in health outcomes. Low enrollment limits the scope for detecting health effects: even heavily subsidized schemes often fail to achieve mass enrollment (Dupas and Jain, 2026). Operational weaknesses, including weak provider contracting and low retention, further constrain program reach in Senegal (Rouyard et al., 2022). National audits similarly document billing irregularities among contracted providers (Cour des Comptes, 2019).

Figure 6. Event study estimates: health outcomes





Note: Event-study estimates using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. Neonatal outcomes are derived from DHS birth histories and estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. Both outcomes are binary. *Abbreviations*: DHS = Demographic and Health Surveys.

5.2 Post-2013 CBHI

In September 2013, Senegal launched a national UHC strategy in which CBHI expansion was a central component. I estimate effects restricting to departments first exposed to CBHI after 2013, when the program received greater government subsidies and technical assistance. Most departments (34 of 45, or 76%) first gained exposure in this period.

Results are shown in Appendix Figure C1. Estimated effects on insurance coverage (5.8 percentage points, 95% CI: 2.3, 9.4) and OOP costs (−19.6 percentage points, 95% CI: −29.7, −9.5) are similar to the full-sample estimates. Effects on health care utilization and health outcomes remain small and not statistically significant, consistent with the main results. Overall, greater government involvement after 2013 expanded coverage but did not lead to larger effects on utilization or health outcomes.

5.3 Robustness checks

5.3.1 Honest difference-in-differences estimates

To assess sensitivity to potential violations of the parallel trends assumption, I implement the honest difference-in-differences method proposed by Rambachan and Roth (2023). This approach bounds post-treatment estimates by the largest deviation observed between adjacent pre-treatment periods.

Figure D1 presents the results. Point estimates are similar to the main results, but confidence intervals are wider. The estimate for insurance coverage is no longer statistically distinguishable from zero once these bounds are imposed. In contrast, the reduction in high OOP costs remains statistically significant.

Effects on utilization and health outcomes are small and not statistically significant, consistent with the baseline results. Overall, these findings suggest that while the first-stage effect is sensitive to modest violations of parallel trends, the evidence on financial protection is more robust.

5.3.2 Placebo estimates

To assess potential threats to identification from compositional changes or time-varying confounders, I estimate placebo regressions using outcomes not plausibly affected by CBHI. These outcomes, drawn from the DHS women's module, include women's and partners' education, employment status, receipt of a fieldworker visit, and self-reported distance to the nearest health facility.

Figure D2 shows that most placebo outcomes do not change following CBHI introduction. The estimate for fieldworker visits is negative and statistically significant, but small in magnitude and close to zero at the upper bound of the confidence interval.

Overall, these results provide little evidence of differential trends in predetermined characteristics and do not suggest that the main findings are driven by compositional changes or confounding shocks.

5.3.3 Robustness to exposure measurement

I conduct three additional robustness checks to address potential concerns about exposure classification. Results are shown in Appendix Figures D3–D5.

Excluding departments with incomplete rollout data. In four departments (Tivaouane, Mbour, Thiès, and Fatick), administrative records do not fully document MHO launch years, which may introduce exposure misclassification. Excluding these departments yields estimates similar to the baseline. Coverage increases by 6.7 percentage points (95% CI: 2.4, 11.1), and the probability of high OOP costs declines by 22.6 percentage points (95% CI: 10.4, 34.8). The effect on secondary or tertiary facility delivery is positive but not statistically significant (2.3 percentage points, 95% CI: −0.5, 5.1). Effects on other outcomes remain close to zero.

Restricting to post-2009 data. DHS cluster displacement protocols changed in 2009, constraining displacement within department boundaries. Earlier surveys may therefore introduce exposure misclassification if clusters are displaced across departments. Restricting the sample to post-2009 rounds yields estimates that are similar in direction to the baseline, though less precisely estimated due to the smaller sample. Effects on utilization and health outcomes remain close to zero. First-stage and cost outcomes cannot be estimated in this subsample due to limited overlap with the SPA data.

Restricting to departments with synchronized rollout. MHOs were often introduced at different times within a department, so the department-level exposure measure may mask within-department variation. I restrict the sample to 10 departments where all MHOs launched in the same year. The ANC visits index increases by 4.3 percentage points (95% CI: 0.3, 8.3), in contrast to the near-zero baseline estimate, though estimates are less precise given the smaller sample. Effects on delivery care and health outcomes remain close to zero. First-stage and cost outcomes cannot be estimated in this subsample due to limited overlap with the SPA data.

Overall, the main findings on insurance coverage and financial protection are robust across these checks. Results on utilization and health outcomes remain broadly consistent with the baseline, with some variation in point estimates that does not alter the main conclusions.

5.3.4 Continuous treatment analysis

I examine whether the binary treatment definition masks variation in CBHI intensity by constructing a continuous measure of the share of each department's population covered by CBHI in a given year. I implement the estimator proposed by de Chaisemartin and D'Haultfoeuille (2024), which extends DiD to settings with variation in treatment intensity over time and addresses concerns related to negative weights and heterogeneous effects. The analysis requires aggregation to the department-year level, yielding a panel of 45 departments observed over time.

Figure D6 presents the results. Estimates are generally less precise than in the main analysis, reflecting the smaller effective sample size after aggregation. The estimate for insurance coverage is negative and not statistically distinguishable from zero (−5.2 percentage points, 95% CI: −18.0, 7.6). The reduction in high OOP costs is also not statistically significant (−15.4 percentage points, 95% CI: −40.0, 9.3). Estimates for utilization and health outcomes remain close to zero. The delivery care index shows no meaningful effect in this specification (−0.8 percentage points, 95% CI: −5.3, 3.8).

Overall, the continuous treatment specification yields estimates that are directionally consistent with limited effects on utilization and health outcomes, but with substantially reduced precision due to aggregation. These results do not alter the main conclusions.

5.4 Heterogeneity analysis

Figure 7 presents ATE estimates separately for urban and rural subpopulations.

Effects on insurance coverage are larger in urban areas (17.7 percentage points, 95% CI: −3.2, 38.6) than in rural areas (−2.6 percentage points, 95% CI: −22.9, 17.6), though neither estimate is statistically significant. The reduction in high OOP costs is larger in rural areas (−23.1 percentage points, 95% CI: −35.6, −10.7) than in urban areas (−19.9 percentage points, 95% CI: −74.2, 34.5), but only the rural estimate is statistically significant.

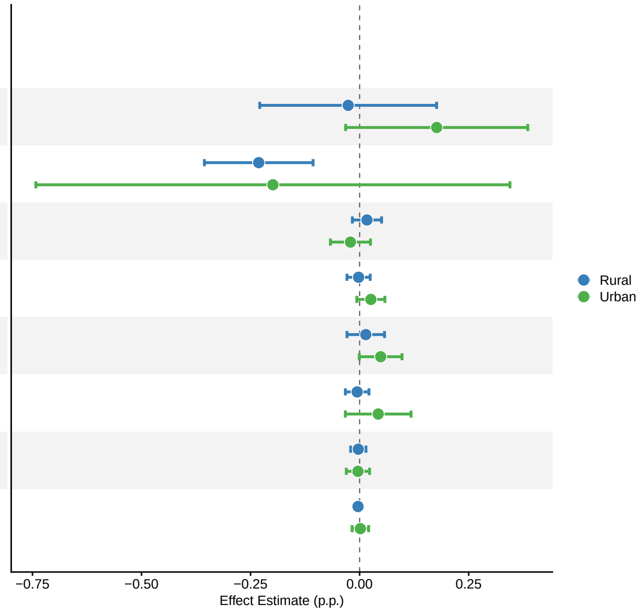
For utilization, the delivery care index increases by 4.8 percentage points in urban areas (95% CI: −0.1, 9.7) and by 1.4 percentage points in rural areas (95% CI: −2.9, 5.7), though neither estimate is statistically significant at conventional levels. The ANC process index increases by 2.6 percentage points in urban areas (95% CI: −0.6, 5.8) and is close to zero in rural areas (−0.2 percentage points, 95% CI: −2.9, 2.4). Effects on ANC visits and secondary or tertiary facility delivery are small and not statistically significant in both strata.

Effects on pregnancy loss and neonatal mortality are close to zero in both urban and rural areas, consistent with the main results.

Overall, the results suggest some differences by place of residence, but the pattern is not uniform across outcomes. Insurance gains appear larger in urban areas, while financial protection effects are more concentrated in rural areas. Across both strata, there is little evidence that these changes translated into improvements in health outcomes.

Figure 7. ATE estimates: urban and rural subgroups

Outcome		Effect (95% CI)	Mean	N
Any Insurance Coverage	Rural	−0.026 (−0.229, 0.176)	0.048	619
	Urban	0.177 (−0.032, 0.386)	0.106	995
OOP Cost in Q5	Rural	−0.231 (−0.356, −0.107)*	0.164	620
	Urban	−0.199 (−0.742, 0.345)	0.211	992
ANC Visits Index	Rural	0.017 (−0.017, 0.050)	0.667	38,200
	Urban	−0.021 (−0.067, 0.025)	0.789	18,640
ANC Process Index	Rural	−0.002 (−0.029, 0.024)	0.723	38,152
	Urban	0.026 (−0.006, 0.058)	0.767	18,619
Delivery Care Index	Rural	0.014 (−0.029, 0.057)	0.762	54,999
	Urban	0.048 (−0.001, 0.097)	0.951	24,682
Secondary/Tertiary Facility Delivery	Rural	−0.006 (−0.033, 0.021)	0.192	54,989
	Urban	0.043 (−0.033, 0.118)	0.491	24,671
Pregnancy Loss	Rural	−0.003 (−0.020, 0.014)	0.094	32,176
	Urban	−0.004 (−0.031, 0.023)	0.112	16,383
Neonatal Death	Rural	−0.004 (−0.012, 0.005)	0.030	196,414
	Urban	0.002 (−0.017, 0.021)	0.021	92,006



Note: This figure presents average treatment effect estimates by urban and rural residence, using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year.

6 Conclusion

This study provides the first nationwide, quasi-experimental evaluation of CBHI in Senegal. I assemble new administrative data on the rollout of MHOs and link them to repeated cross-sectional household and facility surveys spanning nearly three decades. Exploiting staggered implementation across departments, I estimate the causal effects of CBHI exposure on insurance coverage, financial protection, healthcare utilization, and maternal and neonatal health outcomes.

CBHI increased insurance coverage and reduced exposure to high out-of-pocket costs, though the evidence on coverage is sensitive to alternative specifications. Effects on utilization are limited, and I find no measurable impacts on health outcomes. Gains in coverage and financial protection did not translate into consistent improvements in service use or health. These patterns echo the broader CBHI literature, which finds at best modest gains in utilization and financial protection but limited evidence of health improvements (Eze et al., 2023).

They also contrast with larger, government-backed programs in Mexico and China, where insurance expansion was paired with direct supply-side investments and achieved substantially higher enrollment (Conti and Ginja, 2023; Gruber et al., 2023). Voluntary CBHI in Senegal faces structural barriers, including adverse selection, weak administrative capacity, and low retention, that limit its effectiveness as a pathway to UHC (Rouyard et al., 2022).

Several limitations merit attention. Treatment exposure is based on reported MHO launch years, which may be imprecise for older schemes and do not capture phased rollouts or temporary suspensions. Exposure is defined at the department level and cannot capture individual enrollment or service use. Repeated cross-sectional data prevent tracking individuals over time. The analysis focuses on maternal and neonatal outcomes; effects on other conditions remain unexamined. Future research should examine these dimensions and compare community-based schemes with larger-scale models.

The evidence points to the need for reforms beyond voluntary CBHI. Senegal's recent move toward departmental health insurance units (UDAMs) represents a promising shift, with more professional administrative oversight, wider risk pools, and greater contracting leverage with providers. Evaluating UDAMs relative to traditional CBHI is a critical next step.

Achieving UHC will require more than subsidized insurance. Mandatory or bundled schemes integrated with social protection programs may achieve greater financial protection and care-seeking gains (Boutin et al., 2025). Complementary investments in service quality and provider accountability remain essential (Dupas and Jain, 2026; Das and Do, 2023).

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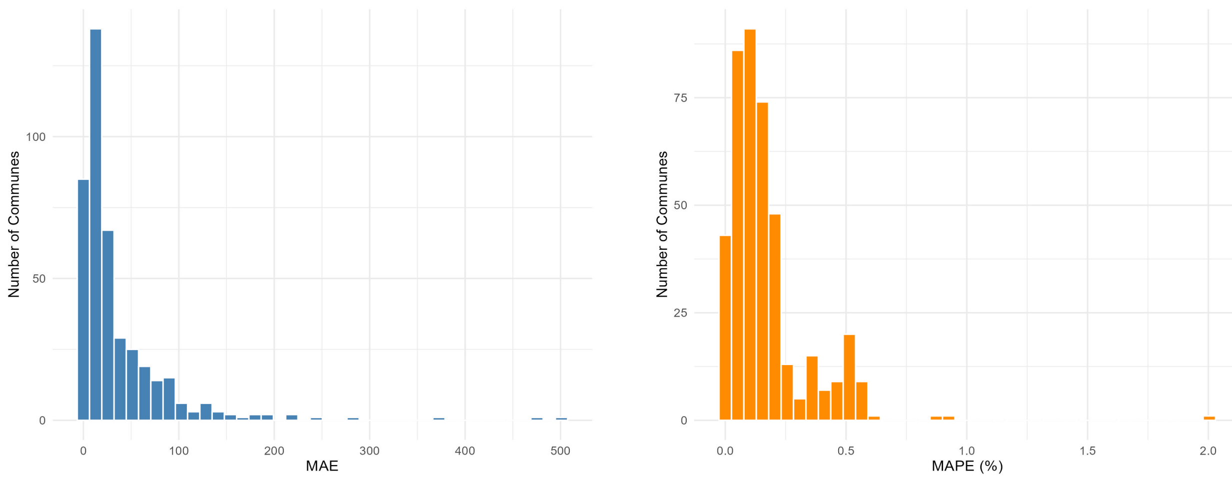
A Appendix A: performance of spline-based population extrapolations

Appendix Figure A1. Performance of population extrapolations

(a) Mean absolute error (MAE) (b) Mean absolute percentage error (MAPE)



Note: This figure summarizes the performance of spline-based population extrapolations using data from the 424 communes that existed in 2002. Panel (a) shows the distribution of mean absolute errors between predicted and observed population values for years with overlap. Panel (b) shows the corresponding distribution of mean absolute percentage errors. The average mean absolute error was 36.2, and the average mean absolute percentage error was 0.17%, indicating high predictive accuracy relative to commune population size.

B Appendix B: additional summary statistics

Appendix Table B1. Summary statistics of the SPA sample, by insurance status

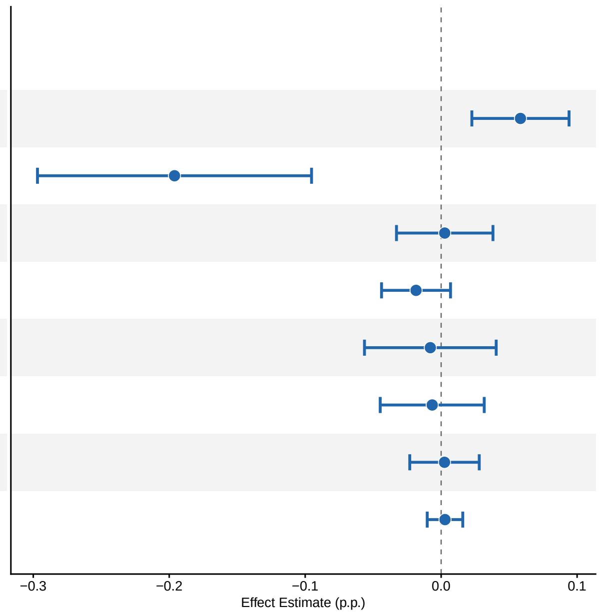
Characteristic	**Overall** N = 5,737	**Uninsured** N = 5,321	**Insured** N = 416
Age	27.83 (7.15)	27.67 (7.14)	29.82 (6.88)
Education level: no education	54%	56%	31%
Education level: primary	27%	26%	30%
Education level: secondary	15%	14%	28%
Education level: higher than secondary	3.9%	3.4%	11%
Location type: urban	57%	56%	78%
Facility type: health post	67%	69%	45%
Facility type: health center	25%	24%	35%
Facility type: hospital	8.2%	7.3%	20%
Facility ownership: public	93%	94%	86%
Facility ownership: non-profit	3.7%	3.6%	4.8%
Facility ownership: private	2.2%	1.7%	8.9%
Facility ownership: faith-based	0.8%	0.8%	0.2%
Service type: antenatal care	45%	46%	42%
Service type: family planning	55%	54%	58%

Note: Characteristics of women interviewed in the SPA exit survey, overall and stratified by insurance status. Age is reported as mean and standard deviation. All other variables are reported as percentages. *Abbreviations*: SPA = Service Provision Assessment.

C Appendix C: subgroup analysis

Appendix Figure C1. ATE estimates: departments exposed to CBHI after 2013

Outcome	Effect (95% CI)	Mean	N
Any Insurance Coverage	0.058 (0.023, 0.094)*	0.091	2,284
OOP Cost in Q5	−0.196 (−0.297, −0.095)*	0.162	2,282
ANC Visits Index	0.003 (−0.033, 0.038)	0.696	33,292
ANC Process Index	−0.018 (−0.044, 0.007)	0.743	33,270
Delivery Care Index	−0.008 (−0.056, 0.040)	0.775	46,786
Secondary/Tertiary Facility Delivery	−0.007 (−0.045, 0.032)	0.295	46,772
Pregnancy Loss	0.002 (−0.023, 0.028)	0.123	40,504
Neonatal Death	0.003 (−0.010, 0.016)	0.031	168,960

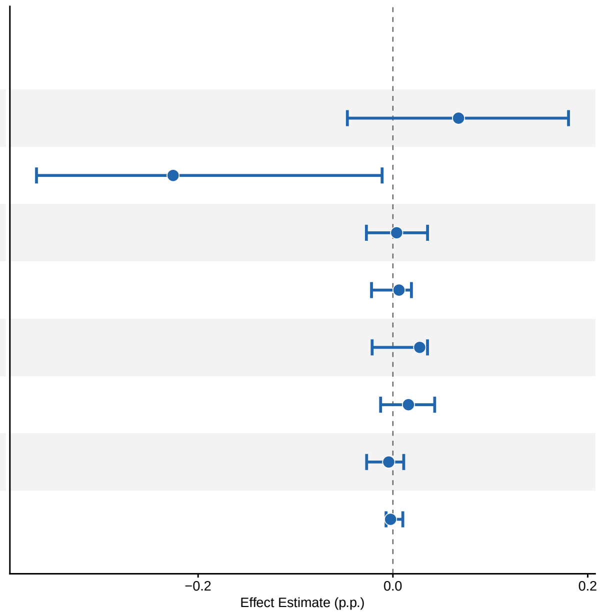


Note: This figure presents average treatment effect estimates restricting to departments first exposed to CBHI after September 2013, when CBHI was adopted as a central component of Senegal's UHC strategy. Estimates use the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; CBHI = community-based health insurance; DHS = Demographic and Health Surveys; FP = family planning; OOP = out-of-pocket; SPA = Service Provision Assessment; UHC = universal health coverage.

D Appendix D: robustness checks

Appendix Figure D1. Honest difference-in-differences estimates

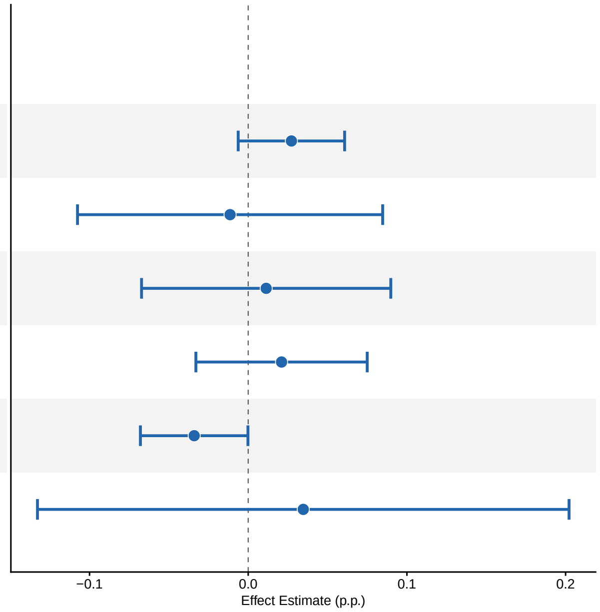
Outcome	Effect (95% CI)	Mean	N
Any Insurance Coverage	0.067 (−0.047, 0.180)	0.076	1,614
OOP Cost in Q5	−0.226 (−0.366, −0.011)*	0.187	1,612
ANC Visits Index	0.004 (−0.027, 0.036)	0.707	56,840
ANC Process Index	0.006 (−0.022, 0.019)	0.737	56,771
Delivery Care Index	0.028 (−0.021, 0.036)	0.820	79,681
Secondary/Tertiary Facility Delivery	0.016 (−0.013, 0.043)	0.284	79,660
Pregnancy Loss	−0.004 (−0.027, 0.011)	0.100	48,559
Neonatal Death	−0.002 (−0.007, 0.010)	0.028	288,420



Note: This figure presents sensitivity bounds using the honest difference-in-differences method proposed by Rambachan and Roth (2023). Bounds are constructed under the assumption that deviations from parallel trends are no larger than the largest fluctuation observed between adjacent pre-treatment periods. All other estimation details follow the baseline specification described in Section 4.

Appendix Figure D2. Placebo estimates

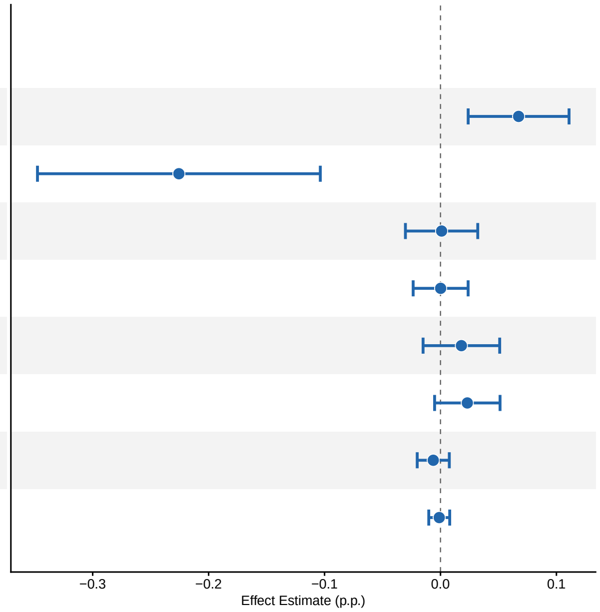
Outcome	Effect (95% CI)	Mean	N
Age: 35+	0.027 (−0.006, 0.061)	0.267	75,803
Education: Primary+	−0.011 (−0.108, 0.085)	0.480	75,803
Husband's Education: Primary+	0.011 (−0.067, 0.090)	0.270	50,385
Employed	0.021 (−0.033, 0.075)	0.453	75,692
Field Worker Visit (Past Year)	−0.034 (−0.068, −0.000)*	0.098	45,287
Care Barrier: Distance to Facility	0.035 (−0.133, 0.202)	0.261	21,014



Note: This figure presents placebo average treatment effect estimates using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. Outcomes are drawn from the DHS women's module and are not expected to be affected by CBHI exposure. *Abbreviations*: CBHI = community-based health insurance; DHS = Demographic and Health Surveys.

Appendix Figure D3. ATE estimates: excluding departments with incomplete rollout data

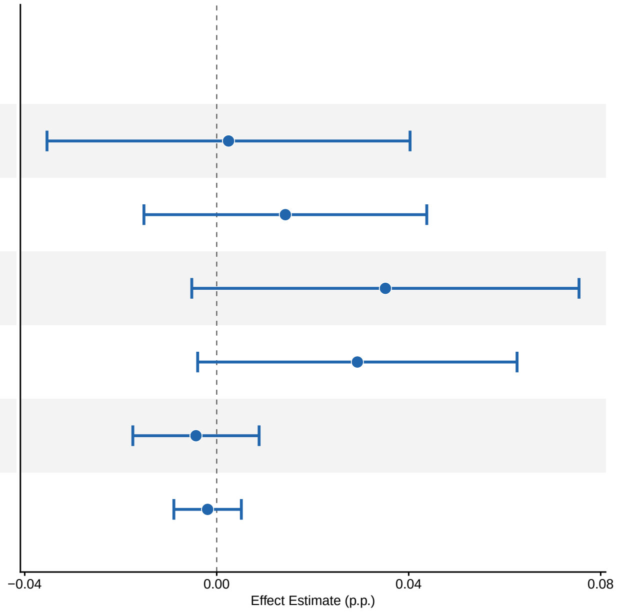
Outcome	Effect (95% CI)	Mean	N
Any Insurance Coverage	0.067 (0.024, 0.111)*	0.084	1,614
OOP Cost in Q5	−0.226 (−0.348, −0.104)*	0.208	1,612
ANC Visits Index	0.001 (−0.030, 0.032)	0.696	49,349
ANC Process Index	0.000 (−0.024, 0.024)	0.730	49,288
Delivery Care Index	0.018 (−0.015, 0.051)	0.796	69,604
Secondary/Tertiary Facility Delivery	0.023 (−0.005, 0.051)	0.277	69,583
Pregnancy Loss	−0.006 (−0.020, 0.008)	0.102	42,964
Neonatal Death	−0.001 (−0.010, 0.008)	0.030	250,834



Note: This sensitivity analysis excludes four departments (Tivaouane, Mbour, Thiès, and Fatick) with incomplete information on MHO launch dates, to reduce potential exposure misclassification. Estimates use the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; DHS = Demographic and Health Surveys; FP = family planning; MHO = mutual health organization; OOP = out-of-pocket; SPA = Service Provision Assessment.

Appendix Figure D4. ATE estimates: restricting to post-2009 observations

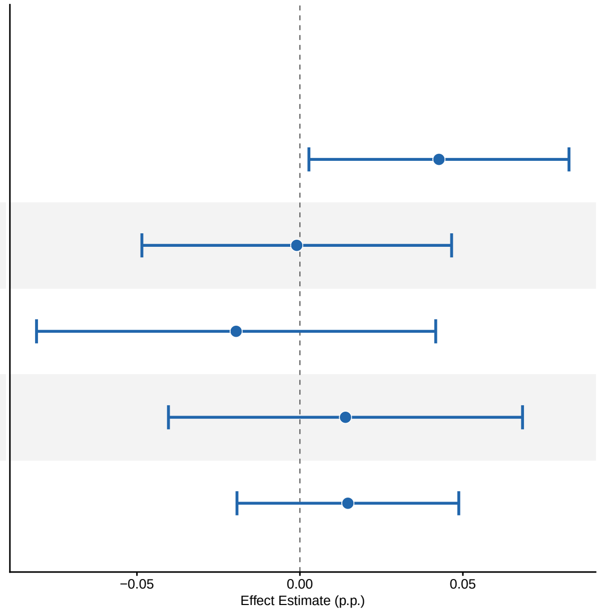
Outcome	Effect (95% CI)	Mean	N
ANC Visits Index	0.002 (−0.035, 0.040)	0.707	27,467
ANC Process Index	0.014 (−0.015, 0.044)	0.737	27,469
Delivery Care Index	0.035 (−0.005, 0.075)	0.820	41,230
Secondary/Tertiary Facility Delivery	0.029 (−0.004, 0.063)	0.284	41,230
Pregnancy Loss	−0.004 (−0.017, 0.009)	0.100	48,559
Neonatal Death	−0.002 (−0.009, 0.005)	0.028	205,102



Note: This sensitivity analysis restricts the sample to post-2009 observations, when DHS cluster displacement was constrained within department boundaries. This reduces exposure misclassification due to GPS displacement. Estimates use the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; DHS = Demographic and Health Surveys; GPS = global positioning system.

Appendix Figure D5. ATE estimates: restricting to departments with synchronized rollout

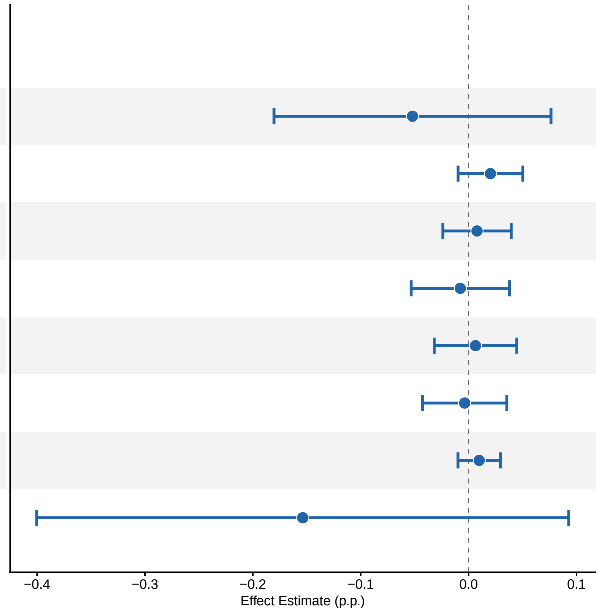
Outcome	Effect (95% CI)	Mean	N
ANC Visits Index	0.043 (0.003, 0.083)*	0.710	7,108
ANC Process Index	−0.001 (−0.048, 0.047)	0.708	7,139
Delivery Care Index	−0.020 (−0.081, 0.042)	0.772	10,479
Secondary/Tertiary Facility Delivery	0.014 (−0.040, 0.068)	0.166	10,473
Neonatal Death	0.015 (−0.019, 0.049)	0.020	37,087



Note: This sensitivity analysis restricts the sample to 10 departments in which all MHOs launched in the same year, to reduce misclassification from within-department variation in rollout timing. Estimates use the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. Standard errors are clustered at the department level. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; DHS = Demographic and Health Surveys; MHO = mutual health organization.

Appendix Figure D6. ATE estimates: continuous treatment

Outcome	Effect (95% CI)	Mean	N
Any Insurance Coverage	−0.052 (−0.180, 0.076)	0.060	69
ANC Visits Index	0.020 (−0.010, 0.050)	0.585	760
ANC Process Index	0.008 (−0.024, 0.039)	0.671	760
Delivery Care Index	−0.008 (−0.053, 0.038)	0.603	760
Secondary/Tertiary Facility Delivery	0.006 (−0.032, 0.045)	0.274	760
Pregnancy Loss	−0.004 (−0.043, 0.035)	0.066	211
Neonatal Death	0.010 (−0.010, 0.030)	0.043	760
OOP Cost in Q5	−0.154 (−0.400, 0.093)	0.150	69



Note: This figure presents average treatment effect estimates using the estimator proposed by de Chaisemartin and D'Haultfoeuille (2024), which allows for continuous treatment intensity and addresses heterogeneity bias in staggered difference-in-differences settings. CBHI exposure is measured as the proportion of each department's population covered by CBHI in a given year. Data are aggregated to the department-year level. Standard errors are clustered at the department level. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. *Abbreviations*: ANC = antenatal care; ATE = average treatment effect; CBHI = community-based health insurance; DHS = Demographic and Health Surveys; FP = family planning; OOP = out-of-pocket.